

Chapter 12 Questions

- Interpret the chemical equation in terms of interacting particles:** $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$
 - Two atoms of hydrogen react with one atom of oxygen to form two formula units of water.
 - Two molecules of hydrogen react with one molecule of oxygen to form two molecules of water.
 - Two moles of hydrogen react with one mole of oxygen to form two moles of water.
 - Two grams of hydrogen react with one gram of oxygen to form two grams of water.
- Interpret the chemical equation in terms of interacting numbers of moles:** $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
 - One molecule of nitrogen reacts with three molecules of hydrogen to form two molecules of ammonia.
 - One gram of nitrogen reacts with three grams of hydrogen to form two grams of ammonia.
 - One mole of nitrogen reacts with three moles of hydrogen to form two moles of ammonia.
 - One atom of nitrogen reacts with three atoms of hydrogen to form two atoms of ammonia.
- Calculate the mass of the reactants and products for the combustion of hydrogen:** $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$. Which statement best demonstrates that the balanced equation obeys the law of conservation of mass?
 - 4.0 g of reactants produces 32.0 g of products.
 - 36.0 g of reactants (4.0 g H_2 + 32.0 g O_2) produces 36.0 g of H_2O .
 - 18.0 g of reactants produces 18.0 g of products.
 - 36.0 g of reactants produces 2.0 g of H_2 and 16.0 g of O_2 .
- Which of the following best explains the term "mole ratio" and when it is used?**
 - A ratio of the molar masses of two substances, used to calculate volume.
 - A conversion factor derived from the coefficients of a balanced chemical equation interpreted in terms of moles.
 - A ratio of the subscripts in a chemical formula, used to determine empirical formulas.
 - A ratio of actual yield to theoretical yield, used to measure efficiency.
- Ammonia is produced by the reaction of nitrogen and hydrogen:** $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$. How many moles of NH_3 form when 6.0 mol of H_2 reacts completely?

A. 2.0 mol	C. 6.0 mol
B. 4.0 mol	D. 9.0 mol
- Ethanol ($\text{C}_2\text{H}_5\text{OH}$) burns in oxygen to produce carbon dioxide and water:** $\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{g})$. How many grams of O_2 are needed to produce 4.00 mol of CO_2 ?

A. 96.0 g	C. 192 g
B. 144 g	D. 288 g
- The combustion of methane produces carbon dioxide and water:** $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$. If you have 48.0 g of CH_4 , how many grams of O_2 are required for a complete reaction?

A. 64.0 g	C. 128 g
B. 96.0 g	D. 192 g
- What information about a chemical reaction is derived from the coefficients in a balanced equation?**
 - The physical states of the reactants and products.
 - The relative number of moles (or particles) of reactants and products.
 - The rate at which the reaction occurs.
 - The amount of energy released or absorbed.
- Sodium metal reacts with water to form aqueous sodium hydroxide and hydrogen gas:** $2\text{Na}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq}) + \text{H}_2(\text{g})$. What mass of water is needed to react completely with 46.0 g of Na?

A. 18.0 g	C. 72.0 g
B. 36.0 g	D. 92.0 g
- Which term describes that portion of chemistry dealing with numerical relationships in chemical reactions; the calculation of quantities of substances involved in chemical equations?**

A. Actual yield	C. Percent yield
B. Stoichiometry	D. Mole ratio

- 11. What is the significance of the limiting reagent in a chemical reaction?**
- A. It determines the maximum amount of product that can be formed.
 - B. It is the reactant that is left over after the reaction stops.
 - C. It speeds up the rate of the chemical reaction.
 - D. It indicates that the theoretical yield will be 100%.
- 12. How would you identify a limiting reagent in a chemical reaction?**
- A. Find the reactant with the lowest starting mass.
 - B. Find the reactant with the lowest coefficient in the balanced equation.
 - C. Express quantities of reactants as moles and compare them to the mole ratios from the balanced equation.
 - D. Express quantities as volumes and find the smallest volume.
- 13. In a reaction chamber, 4.0 mol of nitrogen (N_2) is mixed with 9.0 mol of hydrogen (H_2) to produce ammonia: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$. Identify the limiting reagent and calculate the number of moles of product formed.**
- A. N_2 is limiting; 8.0 mol NH_3 forms.
 - B. H_2 is limiting; 6.0 mol NH_3 forms.
 - C. N_2 is limiting; 4.0 mol NH_3 forms.
 - D. H_2 is limiting; 9.0 mol NH_3 forms.
- 14. Heating calcium carbonate produces calcium oxide and carbon dioxide: $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$. When 50.0 g of CaCO_3 is heated, 22.4 g of CaO is produced. Assuming a theoretical yield of 28.0 g CaO , what is the percent yield of this reaction?**
- A. 44.8%
 - B. 50.0%
 - C. 80.0%
 - D. 89.2%
- 15. What is the amount of product that forms when a reaction is carried out in the laboratory?**
- A. Actual yield
 - B. Theoretical yield
 - C. Limiting reagent
 - D. Excess reagent
- 16. Which of the following is any reactant that is used up first in a chemical reaction?**
- A. Excess reagent
 - B. Solute
 - C. Limiting reagent
 - D. Catalyst
- 17. How is percent yield defined?**
- A. The amount of product that could form during a reaction calculated from a balanced chemical equation.
 - B. A conversion factor derived from the coefficients of a balanced chemical equation.
 - C. The ratio of the actual yield to the theoretical yield for a chemical reaction expressed as a percentage; a measure of the efficiency of a reaction.
 - D. A reagent present in a quantity that is more than sufficient to react with a limiting reagent.
- 18. Which term refers to the amount of product that could form during a reaction calculated from a balanced chemical equation, representing the maximum amount of product that could be formed from a given amount of reactant?**
- A. Actual yield
 - B. Excess reagent
 - C. Percent yield
 - D. Theoretical yield

Instructions: Match the vocabulary term (A-G) with its correct definition (19-25).

- _____ A. Stoichiometry
- _____ B. Mole ratio
- _____ C. Actual yield
- _____ D. Excess reagent
- _____ E. Limiting reagent
- _____ F. Percent yield
- _____ G. Theoretical yield

- 19.** A conversion factor derived from the coefficients of a balanced chemical equation interpreted in terms of moles.
- 20.** The amount of product that forms when a reaction is carried out in the laboratory.
- 21.** The amount of product that could form during a reaction is calculated from a balanced chemical equation; it represents the maximum amount of product that could be formed from a given amount of reactant.
- 22.** Any reactant that is used up first in a chemical reaction; it determines the amount of product that can be formed in the reaction.
- 23.** That portion of chemistry deals with numerical relationships in chemical reactions; the calculation of quantities of substances involved in chemical equations.
- 24.** The ratio of the actual yield to the theoretical yield for a chemical reaction expressed as a percentage; a measure of the efficiency of a reaction.
- 25.** A reagent present in a quantity that is more than sufficient to react with a limiting reagent; any reactant that remains after the limiting reagent is used up in a chemical reaction.

Chapter 14 Questions

- When a sample of gas is compressed, what occurs at the particulate level?**
 - The particles themselves shrink in volume.
 - The space between the particles is reduced.
 - The kinetic energy of the particles decreases.
 - The mass of the gas particles decreases.
- Why does heating a contained gas at a constant volume cause an increase in pressure?**
 - The heat creates more gas particles inside the container.
 - The volume of the container expands slightly to accommodate the heat.
 - The increased kinetic energy of the particles causes more frequent and forceful collisions.
 - The gas molecules break apart into smaller atoms, increasing the number of collisions.
- What happens to the volume of a balloon when it is moved from a warm room to the freezing outdoors, and why?**
 - The volume decreases because the molecules have less kinetic energy and exert less pressure on the inside of the balloon.
 - The volume increases because cold air exerts less external atmospheric pressure.
 - The volume remains the same, but the walls of the balloon become rigid.
 - The volume increases because gas particles naturally expand in colder temperatures.
- A rigid steel tank contains 2.0 moles of oxygen gas. If another 4.0 moles of oxygen gas is added while the temperature and volume remain constant, what happens to the pressure?**
 - The pressure remains the same.
 - The pressure doubles.
 - The pressure triples.
 - The pressure decreases by a third.
- If a gas is compressed from 10.0 L to 2.0 L at a constant temperature, what happens to the pressure?**
 - It becomes one-fifth of the original pressure.
 - It doubles.
 - It quintuples.
 - It is reduced by half.
- Why does gas pressure decrease when gas is slowly removed from a container with a fixed volume?**
 - The remaining particles expand to fill the missing mass.
 - There is a decrease in the number of particle collisions with the container walls.
 - The temperature of the gas automatically drops.
 - The kinetic energy of the remaining particles increases to compensate.
- Which of the following is the correct mathematical equation for Charles's law?**
 - $P_1V_1 = P_2V_2$
 - $V_1/T_1 = V_2/T_2$
 - $P_1/T_1 = P_2/T_2$
 - $PV = nRT$
- The gas in a closed container has a pressure of 4.00×10^2 kPa at 400 K. What will the pressure be if the temperature is lowered to 200 K?**
 - 1.00×10^2 kPa
 - 2.00×10^2 kPa
 - 4.00×10^2 kPa
 - 8.00×10^2 kPa
- Calculate the initial volume of a gas at a pressure of 2.00×10^2 kPa if its final volume is 4.00 L at a pressure of 1.50×10^2 kPa. (Assume constant temperature).**
 - 3.00 L
 - 4.00 L
 - 5.33 L
 - 6.00 L
- A gas with a volume of 2.0 L at 100.0 kPa expands until the pressure drops to 25.0 kPa. What is its new volume if the temperature doesn't change?**
 - 0.50 L
 - 4.0 L
 - 8.0 L
 - 10.0 L

11. A gas with an initial volume of 400 mL at 300 K is heated until its volume reaches 1200 mL. What is the new temperature of the gas if the pressure remains constant?
- A. 100 K
B. 600 K
C. 900 K
D. 1200 K
12. Which of the following represents the combined gas law?
- A. $P_1V_1 = P_2V_2$
B. $V_1/T_1 = V_2/T_2$
C. $(P_1V_1)/T_1 = (P_2V_2)/T_2$
D. $P_{\text{total}} = P_1 + P_2 + P_3$
13. A sealed cylinder contains nitrogen gas at a pressure of 1.50×10^2 kPa and a temperature of 300 K. If the temperature increases to 400 K, what is the new pressure in the cylinder?
- A. 1.00×10^2 kPa
B. 2.00×10^2 kPa
C. 3.00×10^2 kPa
D. 4.50×10^2 kPa
14. How can Gay-Lussac's law be derived from the combined gas law?
- A. By assuming temperature is constant, the T terms cancel out.
B. By assuming volume is constant, so the V_1 and V_2 terms cancel out.
C. By assuming pressure is constant, so the P_1 and P_2 terms cancel out.
D. By assuming the number of moles is constant.
15. Which of the following correctly describes an ideal gas?
- A. Its particles have measurable volume and strong intermolecular attractions.
B. It condenses into a liquid at absolute zero.
C. Its particles have no volume, there are no attractions between them, and collisions are perfectly elastic.
D. It perfectly follows the gas laws only at extremely low temperatures and high pressures.
16. Why is it impossible for a truly ideal gas to exist?
- A. Because all real gas particles have a finite volume and experience attractive forces.
B. Because collisions between gas molecules always result in a loss of total kinetic energy.
C. Because real gases cannot be compressed.
D. Because real gases do not exhibit predictable mathematical behaviors.
17. What is the volume occupied by 2.00 mol of a gas at 300 K if the pressure is 100.0 kPa? ($R = 8.31 \text{ (L}\cdot\text{kPa)/(K}\cdot\text{mol)}$)
- A. 24.9 L
B. 33.2 L
C. 49.9 L
D. 66.4 L
18. What volume will 14.0 g of nitrogen gas (N_2) occupy at 300 K and a pressure of 50.0 kPa?
- A. 12.5 L
B. 24.9 L
C. 49.9 L
D. 83.1 L
19. If 8.00 g of helium gas (He) is in a 4.00-L container at 400 K, what is the pressure in the container?
- A. 8.31×10^2 kPa
B. 1.66×10^3 kPa
C. 3.32×10^3 kPa
D. 6.64×10^3 kPa
20. A balloon has a volume of 1.00×10^2 L at 200.0 kPa pressure and a temperature of 300 K. What is the approximate mass of argon gas (Ar) inside the balloon?
- a. 8.0×10^1 g
b. 1.6×10^2 g
c. 3.2×10^2 g
d. 6.4×10^2 g
21. Which statement best describes Dalton's law of partial pressures?
- A. The volume of a mixture of gases equals the sum of the individual volumes.
B. At constant volume and temperature, the total pressure of a gas mixture is equal to the sum of the partial pressures of the component gases.
C. The rate of effusion of a gas is inversely proportional to its molar mass.
D. Gas pressure is directly proportional to temperature in a rigid container.

22. Which gas effuses faster, helium (He) or neon (Ne), and by approximately what factor?
- A. Helium effuses faster by a factor of 2.25.
 - B. Neon effuses faster by a factor of 2.25.
 - C. Helium effuses faster by a factor of 5.0.
 - D. Neon effuses faster by a factor of 5.0.
23. Which gas will effuse faster at the same temperature: carbon dioxide (CO₂) or molecular nitrogen (N₂)?
- A. Carbon dioxide, because it has a higher molar mass.
 - B. Molecular nitrogen, because it has a lower molar mass.
 - C. They will effuse at the same rate.
 - D. It depends on the size of the container.
24. Calculate the approximate ratio of the velocity of hydrogen molecules (H₂) to the velocity of oxygen molecules (O₂) at the same temperature.
- A. 1:4
 - B. 2:1
 - C. 4:1
 - D. 16:1
25. Calculate the approximate ratio of the velocity of helium atoms (He) to the velocity of argon atoms (Ar) at the same temperature.
- A. 3.16:1
 - B. 1:3.16
 - C. 10:1
 - D. 1:10
26. Which term describes the measure of how much the volume of matter decreases under pressure?
- A. Diffusion
 - B. Compressibility
 - C. Effusion
 - D. Partial pressure
27. Which law states that for a given mass of gas at constant temperature, the volume varies inversely with pressure?
- A. Charles's law
 - B. Gay-Lussac's law
 - C. Boyle's law
 - D. Graham's law of effusion
28. Which law states that the volume of a fixed mass of gas is directly proportional to its Kelvin temperature if the pressure is kept constant?
- A. Boyle's law
 - B. Charles's law
 - C. Dalton's law
 - D. Combined gas law
29. What do we call the relationship $PV = nRT$ that describes the behavior of an ideal gas?
- A. Combined gas law
 - B. Gas law (ideal gas law)
 - C. Dalton's law of partial pressures
 - D. Gas constant
30. Which term defines the tendency of molecules to move toward areas of lower concentration until the concentration is uniform throughout?
- A. Compressibility
 - B. Diffusion
 - C. Effusion
 - D. Partial pressure
31. Which term defines the process that occurs when a gas escapes through a tiny hole in its container?
- A. Diffusion
 - B. Compressibility
 - C. Evaporation
 - D. Effusion
32. Which law states that the rate of effusion of a gas is inversely proportional to the square root of its molar mass?
- A. Graham's law of effusion
 - B. Dalton's law of partial pressures
 - C. Boyle's law
 - D. Gay-Lussac's law
33. What is the term for the contribution each individual gas makes to the total pressure of a mixture?
- A. Compressibility
 - B. Gas constant
 - C. Partial pressure
 - D. Diffusion

Match each term to its correct definition.

Terms

- 34. _____ Compressibility
- 35. _____ Boyle's law
- 36. _____ Charles's law
- 37. _____ Combined gas law
- 38. _____ Gay-Lussac's law
- 39. _____ Gas constant
- 40. _____ Gas law
- 41. _____ Dalton's law of partial pressures
- 42. _____ Diffusion
- 43. _____ Effusion
- 44. _____ Graham's law of effusion
- 45. _____ Partial pressure

Definitions

- A. A measure of how much the volume of matter decreases under pressure.
- B. For a given mass of gas at constant temperature, the volume of the gas varies inversely with pressure.
- C. The volume of a fixed mass of gas is directly proportional to its Kelvin temperature if the pressure is kept constant.
- D. The law that describes the relationship among the pressure, temperature, and volume of an enclosed gas.
- E. The pressure of a gas is directly proportional to the Kelvin temperature if the volume is constant.
- F. The constant with the symbol R and the value $8.31 \text{ (L}\cdot\text{kPa)/(K}\cdot\text{mol)}$.
- G. The relationship $PV = nRT$ which describes the behavior of an ideal gas.
- H. At constant volume and temperature, the total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of the component gases.
- I. The tendency of molecules to move toward areas of lower concentration until the concentration is uniform throughout.
- J. The process that occurs when a gas escapes through a tiny hole in its container.
- K. The rate of effusion of a gas is inversely proportional to the square root of its molar mass.
- L. The contribution each gas in a mixture of gases makes to the total pressure.

Chapter 15 Questions

- What primarily causes the inward pull that minimizes the surface area of water?**
 - Metallic bonding
 - Strong hydrogen bonding
 - Ionic attractions
 - Dispersion forces
- Why is the behavior of surface water molecules different from those deep within the liquid?**
 - They form ionic bonds with the air.
 - They possess a higher kinetic energy.
 - Surface molecules are attracted to the liquid molecules below but not to the air above.
 - They are repelled by the liquid molecules below them.
- Which of the following is an observable effect of water's surface tension?**
 - Water boils at 100°C.
 - Drops are spherical, and objects denser than liquid can float on its surface.
 - Ice dissolves in liquid water.
 - Sugar dissolves rapidly in water.
- How does a wetting agent, like soap, act as a surfactant?**
 - It creates a stable suspension.
 - It increases the boiling point of the solution.
 - It increases surface tension by strengthening bonds.
 - It interferes with hydrogen bonding between water molecules and reduces surface tension.
- The low tendency for water molecules to escape from the liquid phase explains its unusually low vapor pressure. What causes this?**
 - High molar mass
 - Atmospheric pressure
 - Hydrogen bonds holding water molecules to each other
 - Covalent bonds between oxygen and hydrogen
- Why do large bodies of water, like oceans and lakes, evaporate relatively slowly?**
 - High atmospheric pressure
 - High surface tension
 - High salinity levels
 - Water has a low vapor pressure.
- How does the molecular framework of ice compare to that of liquid water?**
 - Ice is a regular open framework of hydrogen-bonded water molecules arranged like a honeycomb.
 - Ice has no hydrogen bonds between its molecules.
 - Ice is a highly compressed, unstructured solid.
 - Liquid water has a more rigid crystalline structure than ice.
- If the solid form of water were denser than its liquid form, what would be a major consequence?**
 - Water would evaporate instantly.
 - Ice would melt at a higher temperature.
 - Oceans would boil.
 - Bodies of water would freeze from the bottom up, killing many forms of aquatic life.
- What is the specific role of hydrogen bonds in solid ice?**
 - They force the water molecules into a dense, irregular shape.
 - They break apart the water molecules.
 - They hold the molecules in a regular, open structure.
 - They increase the overall temperature of the ice.
- What specific characteristic defines an aqueous solution?**
 - It is a concentrated mixture.
 - It conducts electricity purely as a liquid.
 - It contains a dissolved gas.
 - It is a solution that has water as the solvent.

11. **When mixing table sugar into water to create a solution, what are the respective roles of each substance?**
- A. Both are solutes.
 - B. Water is the solvent; sugar is the solute.
 - C. Sugar is the solvent; water is the solute.
 - D. Both are solvents.
12. **Why do nonpolar compounds generally fail to dissolve in water?**
- A. They form strong hydrogen bonds with water.
 - B. Polar water molecules electrostatically attract ions and polar molecules, but nonpolar compounds are unaffected because they have no charges.
 - C. They vaporize upon contact with water.
 - D. They are too large to fit between water molecules.
13. **Why is filtration an ineffective method for separating dissolved salt or sugar from water?**
- A. The molecules and ions are smaller than the pores of the filter.
 - B. The water dissolves the filter paper.
 - C. The salt and sugar bond chemically to the filter.
 - D. Solutions are too viscous to pass through a filter.
14. **What occurs during the process of solvation?**
- A. The solvent chemically reacts with the solute.
 - B. Solvent molecules evaporate quickly.
 - C. Cations and anions become surrounded by solvent molecules.
 - D. Cations and anions precipitate out of the solution.
15. **Based on the rule of "like dissolves like," which of the following substances will NOT dissolve appreciably in water?**
- A. HCl (polar)
 - B. NaI (ionic)
 - C. NH₃ (polar)
 - D. CH₄ (nonpolar)
16. **Why does gasoline remain separated when mixed with water?**
- A. Gasoline undergoes a chemical reaction with water.
 - B. Water is polar, and gasoline is nonpolar.
 - C. Gasoline is too dense to mix.
 - D. Water is nonpolar, and gasoline is polar.
17. **For a solution to conduct an electric current, what must be present?**
- A. Nonpolar molecules
 - B. Suspended particles
 - C. Unbonded electrons
 - D. Ions
18. **Why is molten sodium chloride able to conduct electricity?**
- A. It produces free electrons.
 - B. Its ions are free to move toward an electrode.
 - C. It vaporizes easily.
 - D. It contains metallic bonds.
19. **What differentiates an aqueous solution of a strong electrolyte from that of a weak electrolyte?**
- A. A strong electrolyte only partially dissolves.
 - B. A strong electrolyte forms a precipitate.
 - C. A strong electrolyte doesn't conduct electricity.
 - D. A strong electrolyte is almost totally ionized.

20. What does a substance's "water of hydration" refer to?
- A. Water that has been purified for lab use.
 - B. Water used to dissolve an electrolyte.
 - C. Water in the crystal structure of a substance.
 - D. Water that forms on the outside of a cold glass.
21. What is the correct chemical formula for barium hydroxide octahydrate?
- A. $\text{BaO} \cdot 8\text{H}_2\text{O}$
 - B. $\text{BaH}_2 \cdot 8\text{H}_2\text{O}$
 - C. $\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$
 - D. $\text{Ba}(\text{OH})_2 \cdot 6\text{H}_2\text{O}$
22. What is the correct name for the hydrate $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$?
- A. Iron sulfate hexahydrate
 - B. Iron(III) sulfate heptahydrate
 - C. Iron(II) sulfide heptahydrate
 - D. Iron(II) sulfate heptahydrate
23. When Epsom salt ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$) is heated to 150°C , it changes to the monohydrate form. Which equation correctly represents this change?
- A. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O} \rightarrow \text{MgSO}_4 \cdot \text{H}_2\text{O} + 6\text{H}_2\text{O}$
 - B. $\text{MgSO}_4 \cdot \text{H}_2\text{O} + 6\text{H}_2\text{O} \rightarrow \text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
 - C. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O} \rightarrow \text{MgSO}_4 + 7\text{H}_2\text{O}$
 - D. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O} \rightarrow \text{Mg} + \text{SO}_4 + 7\text{H}_2\text{O}$
24. How do hygroscopic substances function effectively as desiccants?
- A. They repel water molecules.
 - B. They freeze ambient water.
 - C. They release water vapor into the air.
 - D. They absorb water vapor from the air.
25. Why must hygroscopic compounds be stored in tightly sealed containers?
- A. They decompose in sunlight.
 - B. They absorb water vapor from air.
 - C. They are highly volatile.
 - D. They react violently with nitrogen.
26. Under what condition will an efflorescent hydrate lose its water of hydration?
- A. When the hydrate has a higher vapor pressure than that of the water vapor in air.
 - B. When it is cooled below freezing.
 - C. When it is dissolved in water.
 - D. When it is placed in a high-humidity environment.
27. What is a key observable difference between a suspension and a solution?
- A. Solutions cannot be filtered.
 - B. Suspensions are always clear.
 - C. A suspension has large particles that settle to the bottom, whereas particles in a solution stay suspended.
 - D. A solution will settle over time, while a suspension will not.
28. Which list correctly orders mixtures in order of increasing particle size?
- A. Solutions, suspensions, colloids
 - B. Colloids, solutions, suspensions
 - C. Solutions, colloids, suspensions
 - D. Suspensions, colloids, solutions
29. What is the Tyndall effect?
- A. The rapid random movement of particles
 - B. The scattering of visible light by colloids and suspensions

- C. The absorption of light by a solution
 - D. The settling of particles in a suspension
30. **Why do true solutions fail to demonstrate the Tyndall effect?**
- A. They absorb all visible light.
 - B. They are completely opaque.
 - C. The solvent particles are too large.
 - D. The molecules or ions are too small to have reflective surfaces.
31. **Brownian motion is observed in colloidal dispersions. What is the cause of this motion?**
- A. Magnetic fields acting on ions
 - B. Collisions of the molecules of the dispersing medium with the small, dispersed colloidal particles
 - C. The Tyndall effect
 - D. Gravitational pull on large particles
32. **What two circumstances help keep colloidal particles in suspension?**
- A. High density and low temperature
 - B. Gravity and air pressure
 - C. Solvation and efflorescence
 - D. Brownian motion and repulsion between like-charged ions absorbed on the surface of the colloidal particles
33. **How can a stable colloidal dispersion be destroyed?**
- A. Add ions with a charge opposite that of the colloidal particles.
 - B. Boil the mixture.
 - C. Add an emulsifier.
 - D. Shine a light through it to trigger the Tyndall effect.
34. **What can be added to a colloidal dispersion to make it stable?**
- A. A desiccant
 - B. An emulsifier
 - C. A strong acid
 - D. An electrolyte
37. **Which term describes a wetting agent that interferes with hydrogen bonding between water molecules and reduces surface tension?**
- A. Desiccant
 - B. Surfactant
 - C. Solute
 - D. Emulsifier
38. **Which term describes the process where cations and anions become surrounded by solvent molecules?**
- A. Efflorescence
 - B. Solvation
 - C. Brownian motion
 - D. Suspension
39. **What do we call a substance that is almost totally ionized in an aqueous solution?**
- A. A strong electrolyte
 - B. A weak electrolyte
 - C. A nonpolar solvent
 - D. A colloid
40. **Which term identifies the phenomenon involving the scattering of visible light by colloids and suspensions?**
- A. Efflorescence
 - B. Solvation
 - C. Tyndall effect
 - D. Brownian motion
41. **What term refers to a hydrate that loses its water of hydration because it has a higher vapor pressure**

than the water vapor in the air?

- A. Hygroscopic
- B. Aqueous
- C. Emulsifier
- D. Efflorescent

Match the term with its definition

- 1. ____ Aqueous solution
- 2. ____ Solvent
- 3. ____ Solute
- 4. ____ Water of hydration
- 5. ____ Desiccant
- 6. ____ Suspension
- 7. ____ Colloid
- 8. ____ Brownian motion
- 9. ____ Emulsifier
- 10. ____ Surfactant

- A. A mixture that has large particles that settle to the bottom.
- B. The dissolving medium in a solution (e.g., water in a sugar-water mixture).
- C. The addition of this substance makes a colloidal dispersion stable.
- D. A solution that has water as the solvent.
- E. Water contained within the crystal structure of a substance.
- F. A wetting agent such as soap or detergent that reduces surface tension.
- G. The substance being dissolved in a solution (e.g., sugar in a sugar-water mixture).
- H. A mixture with a particle size intermediate between solutions and suspensions.
- I. A hygroscopic substance used to absorb water vapor from the air.
- J. Caused by collisions of the molecules of the dispersing medium with the small, dispersed colloidal particles.

Chapter 16

- 1. Name and distinguish between the two components of a solution.**
 - A) The solvent is dissolved in the solute.
 - B) The solute is the substance in which the solvent is dissolved.
 - C) The solvent is the substance in which the solute is dissolved.
 - D) The solute and solvent are immiscible liquids.
- 2. Explain why the dissolved component does not settle out of a solution.**
 - A) Gravity has no effect on dissolved particles.
 - B) Random collisions of the solvent molecules with the solute particles provide enough force to overcome gravity.
 - C) The solute particles are chemically bonded to the container.
 - D) The solute has a lower density than the solvent.
- 3. If a saturated solution of potassium chloride is cooled, what change might you observe?**
 - A) The solution begins to boil.
 - B) The solvent evaporates instantly.
 - C) The solution becomes unsaturated.
 - D) Particles of solute crystallize.
- 4. Can a solution with undissolved solute be supersaturated?**
 - A) Yes, because the presence of solid means it holds maximum capacity.
 - B) Yes, undissolved solute is the definition of supersaturation.
 - C) No; if there were undissolved solute, the excess solute would come out of a supersaturated solution.
 - D) No, because supersaturated solutions only exist at extremely low temperatures.
- 5. What mass of AgNO_3 can be dissolved in 500 g of water at 20°C ?**
 - A) 224 g
 - B) 560 g
 - C) 1.12×10^3 g
 - D) 2.24×10^3 g
- 6. What is the effect of pressure on the solubility of gases in liquids?**
 - A) Solubility decreases with pressure.
 - B) Solubility increases with pressure.
 - C) Pressure has no effect on solubility.
 - D) Solubility fluctuates randomly with pressure changes.
- 7. The solubility of a gas in water at 20°C and 1.00 atm is 0.040 g/L. If the temperature remains constant, what will be the solubility of this gas at 0.50 atm?**
 - A) 0.010 g/L
 - B) 0.020 g/L
 - C) 0.040 g/L
 - D) 0.080 g/L
- 8. Knowing the molarity of a solution is more meaningful than knowing whether a solution is dilute or concentrated. Explain.**
 - A) Dilute and concentrated are exact quantitative terms.
 - B) Molarity indicates whether a solution is saturated or not.
 - C) Dilute and concentrated are relative terms and are not quantitative; molarity provides the exact number of moles of solute per liter of solution.
 - D) Molarity only measures the volume of the solvent.
- 9. Calculate the molarity of a solution containing 2.0 mol of NaCl in 500 mL of solution.**
 - A) 1.0M
 - B) 2.0M
 - C) 4.0M
 - D) 0.25M

10. How many milliliters of a 2.00M HCl solution would you need to dilute to make 250.0 mL of 0.500M HCl?
- A) 15.0 mL
 - B) 31.3 mL
 - C) 62.5 mL
 - D) 125.0 mL
11. Calculate the mass of solute in 2.0 L of a 0.25M NaOH solution. (Assuming a molar mass of approximately 40 g/mol).
- A) 10 g
 - B) 20 g
 - C) 40 g
 - D) 80 g
12. Calculate the grams of solute required to make 1500 g of a 2.0% (m/m) KCl solution.
- A) 15 g
 - B) 30 g
 - C) 75 g
 - D) 300 g
13. What is the concentration (in % v/v) if 40 mL of methanol is diluted with water to a total volume of 200 mL?
- A) 16% (v/v)
 - B) 20% (v/v)
 - C) 25% (v/v)
 - D) 80% (v/v)
14. What are colligative properties?
- A) Properties that depend only on the identity of the solute particles.
 - B) Properties of a solution that depend only on the number of solute particles.
 - C) Properties that depend only on the temperature of the solvent.
 - D) Properties that determine the chemical reactivity of the solution.
15. Which solution has the higher boiling point: 2.0M NaCl or 1.0M MgCl₂?
- A) 2.0M NaCl
 - B) 1.0M MgCl₂
 - C) They have the same boiling point.
 - D) Distilled water has a higher boiling point than both.
16. Why does a 1m solution of CaCl₂ have a lower freezing point than a 1m solution of KCl?
- A) CaCl₂ is more massive than KCl.
 - B) The effective molality of the CaCl₂ solution is 3m, while the effective molality of the KCl solution is 2m.
 - C) Potassium is more reactive than calcium.
 - D) The effective molality of the KCl solution is higher.
17. Explain how a decrease in the vapor pressure of a solution affects its boiling point.
- A) It decreases the boiling point relative to pure solvent.
 - B) When vapor pressure is lowered relative to pure solvent, more energy must be supplied to reach the boiling point; thus the boiling point is increased.
 - C) It has no effect on the boiling point.
 - D) It causes the solution to freeze instead of boil.
18. In cold climates, salt is spread on roads during winter storms. What is the purpose of the salt?
- A) The salt raises the freezing point of the ice.
 - B) The salt lowers the freezing point of the ice-water cooling mixture.
 - C) The salt increases the friction of the road surface only.
 - D) The salt absorbs the water to form a solid hydrate.
19. Distinguish between a 1M solution and a 1m solution.

- A) 1M is 1 mol of solute in 1 L of solution; 1m is 1 mol of solute in 1000 g of solvent.
- B) 1M is 1 mol of solute in 1000 g of solvent; 1m is 1 mol of solute in 1 L of solution.
- C) 1M relies on temperature, while 1m relies on pressure.
- D) There is no structural difference; they are interchangeable terms.

42. Which term describes liquids that are insoluble in one another, such as oil and water?

- A) Miscible
- B) Immiscible
- C) Unsaturated
- D) Solvated

43. The amount of a substance that dissolves in a given quantity of solvent at specified conditions of temperature and pressure to produce a saturated solution is called its:

- A) Molarity
- B) Molality
- C) Solubility
- D) Concentration

44. At a given temperature, the solubility of a gas in a liquid is directly proportional to the pressure of the gas above the liquid. This is known as:

- A) Boyle's Law
- B) Charles's Law
- C) Graham's Law
- D) Henry's law

45. Which term describes a solution containing the maximum amount of solute for a given amount of solvent at a constant temperature and pressure?

- A) Saturated solution
- B) Unsaturated solution
- C) Supersaturated solution
- D) Dilute solution

46. The ratio of the moles of solute in solution to the total number of moles of both solvent and solute is known as the:

- A) Molality
- B) Mole fraction
- C) Molarity
- D) Concentration

Match the vocabulary term on the left with the correct definition on the right.

1. ____ Concentrated solution
2. ____ Dilute solution
3. ____ Supersaturated solution
4. ____ Boiling-point elevation
5. ____ Molality (m)
6. ____ Molal freezing-point depression constant (Kf)
7. ____ Colligative property

- A) A property of a solution that depends only upon the number of solute particles, and not upon their identities.
- B) A solution containing a large amount of solute.
- C) The change in freezing point for a 1-molal solution of a nonvolatile molecular solute.
- D) A solution that contains more solute than it can theoretically hold at a given temperature; excess solute precipitates if a seed crystal is added.
- E) The difference in temperature between the boiling point of a solution and the boiling point of the pure solvent.
- F) The concentration of solute in a solution expressed as the number of moles of solute dissolved in 1 kilogram (1000 g) of solvent.
- G) solution that contains a small amount of solute.

Chapter 17

- Which of the following best describes the law of conservation of energy?**
 - Energy naturally flows from a colder object to a warmer object.
 - In any chemical or physical process, energy is neither created nor destroyed.
 - The potential energy of a system always decreases during a chemical reaction.
 - Energy is only conserved in exothermic reactions, but not in endothermic reactions.
- When a warm block of aluminum is placed in a cool cup of water, what always happens?**
 - Heat flows from the water to the aluminum until they reach the same temperature.
 - Coldness flows from the water into the aluminum block.
 - Heat flows from the aluminum to the water.
 - The temperature of both objects immediately drops to the surrounding room temperature.
- In terms of chemistry, how is potential energy defined?**
 - The energy an object possesses due to its motion.
 - The energy stored in a substance because of its chemical composition.
 - The heat transferred between the system and the surroundings.
 - The energy required to raise the temperature of 1 gram of water by 1°C.
- What two factors primarily determine the heat capacity of an object?**
 - The volume of the substance and its temperature.
 - The chemical composition of the substance and its mass.
 - The atmospheric pressure and the specific heat.
 - The phase of the substance and its density.
- Which of the following statements correctly describes the relationship between a chemistry calorie (cal) and a dietary Calorie (Cal)?**
 - 1 Cal = 100 cal
 - 1000 Cal = 1 cal
 - 1 Cal = 1000 cal
 - 1 Cal = 4.184 cal
- What is the equivalent of 350 cal in joules? (Note: 1 cal = 4.184 J)**
 - 83.6 J
 - 1464 J
 - 1.46×10^3 J
 - 8.36×10^1 J
- Why is it important to define the system and the surroundings in thermochemistry?**
 - To determine the exact mass of the products formed.
 - To accurately measure the heat flow across the boundary between them.
 - Because the surroundings always have a higher temperature than the system.
 - So that the law of conservation of mass can be proven.
- What is the correct sign convention used in thermochemical calculations?**
 - A positive sign is given to heat flow from the system to the surroundings.
 - A negative sign indicates an endothermic process.
 - A negative sign is given to heat flow from the system to the surroundings, and a positive sign indicates heat flow to the system.
 - Both exothermic and endothermic processes use positive signs if the temperature increases.
- Two chemical substances are mixed in a beaker, and the beaker quickly feels freezing cold to the touch. What is the nature of this reaction, and what constitutes the surroundings?**
 - Exothermic; the chemicals are the surroundings.
 - Endothermic; the glass beaker and the surrounding air are the surroundings.
 - Endothermic; only the chemicals inside the beaker are the surroundings.
 - Exothermic; the surrounding air is the system.
- Which of the following processes is considered endothermic?**
 - Condensing water vapor
 - Evaporating liquid nitrogen
 - Burning a piece of wood
 - Freezing water into ice

11. What name is given to a heat change that occurs at constant pressure?
- A) Specific heat
 - B) Entropy
 - C) Enthalpy
 - D) Kinetic energy
12. What is the primary function of a calorimeter?
- A) To measure atmospheric pressure changes.
 - B) To measure enthalpy changes in physical or chemical processes.
 - C) To determine the molecular weight of an unknown compound.
 - D) To accelerate the rate of a chemical reaction.
13. Which of the following is an obvious source of error when using a foam cup as a calorimeter?
- A) The foam cup perfectly reflects all heat back into the solution.
 - B) The reactants may not be completely mixed, and heat can be lost to the air.
 - C) Foam cups generate their own heat, skewing the data.
 - D) The foam cup increases the activation energy of the reaction.
14. What device is specifically used to measure the heat released during a chemical reaction at constant volume?
- A) A standard foam cup calorimeter
 - B) A barometer
 - C) A bomb calorimeter
 - D) A spectrophotometer
15. What are the standard conditions for measuring the heat of combustion?
- A) 101.3 kPa (1 atm) and 0°C
 - B) 101.3 kPa (1 atm) and 25°C
 - C) 100.0 kPa and 100°C
 - D) 0.0 kPa and 25°C
16. What specific information must be provided in a thermochemical equation?
- A) Only the balanced chemical formulas of the reactants and products.
 - B) The molar masses of all substances involved.
 - C) The amount of heat released or absorbed in the chemical change at constant pressure.
 - D) The rate at which the reaction occurs.
17. Why does the temperature of a glass of ice water remain at 0°C even as heat continuously flows from the warmer room into the glass?
- A) Heat is being used to melt the ice rather than increase the kinetic energy of the particles.
 - B) The water is evaporating rapidly, causing a cooling effect.
 - C) Ice has an infinite heat capacity.
 - D) The glass reflects the heat away from the system.
18. Calculate the quantity of heat lost when 2.00 mol of water freezes at 0°C. (Assume the molar heat of fusion for water is 6.01 kJ/mol).
- A) +12.0 kJ
 - B) -12.0 kJ
 - C) -3.01 kJ
 - D) +3.01 kJ
19. Ammonium nitrate dissolves readily in water according to the following equation: $\text{NH}_4\text{NO}_3(\text{s}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{NO}_3^-(\text{aq})$ $\Delta H = 25.7 \text{ kJ/mol}$. Would this process increase or decrease the temperature of the water?
- A) Increase, because the reaction is exothermic.
 - B) Increase, because the reaction is endothermic.
 - C) Decrease, because the reaction is exothermic.
 - D) Decrease, because the reaction is endothermic.
20. Which of the following statements best explains Hess's law of heat summation?
- A) The total enthalpy of a system is equal to its heat capacity multiplied by its mass.
 - B) The enthalpy change of a reaction can be calculated from the known enthalpy changes of two or more other reactions.
 - C) Heat always flows from a higher temperature to a lower temperature.
 - D) The heat of a reaction is only dependent on the atmospheric pressure.
21. The decomposition of nitrogen dioxide requires heat: $2\text{NO}_2(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2\text{O}_2(\text{g})$ $\Delta H = +66.4 \text{ kJ}$. What is the enthalpy change for the formation of 1 mol of NO_2 from its elements, and is this

formation exothermic or endothermic?

- A) +66.4 kJ, endothermic
- B) -33.2 kJ, exothermic

- C) -66.4 kJ, exothermic
- D) +33.2 kJ, endothermic

22. Calculate the enthalpy change for the formation of substance $\text{XY}_4(\text{l})$ from $\text{XY}_2(\text{s})$ and $\text{Y}_2(\text{g})$, given the following thermochemical equations:



Reaction to solve: $\text{XY}_2(\text{s}) + \text{Y}_2(\text{g}) \rightarrow \text{XY}_4(\text{l})$

- A) -650.0 kJ
- B) -150.0 kJ

- C) +150.0 kJ
- D) +650.0 kJ

23. What is the standard heat of formation (ΔH_f°) of a free element in its standard state?

- A) Always a negative value
- B) Always a positive value
- C) Exactly zero
- D) 101.3 kJ

24. Consider the statement, "the more negative the value of ΔH_f° , the more stable the compound." Is this statement generally true or false, and why?

- A) False, because stability implies a higher energy state.
- B) True, because a greater release of heat indicates the compound is more stable relative to its elements.
- C) False, because negative values indicate the compound will rapidly decompose.
- D) True, because negative values mean the compound requires energy to form.

25. Which term describes the amount of heat needed to increase the temperature of exactly 1 g of a substance by 1°C ?

- A) Heat capacity
- B) Specific heat

- C) Enthalpy
- D) Chemical potential energy

26. The study of energy changes that occur during chemical reactions and changes in state is called:

- A) Calorimetry
- B) Stoichiometry

- C) Thermochemistry
- D) Thermodynamics

27. What is the definition of "enthalpy (H)"?

- A) The heat content of a system at constant pressure.
- B) The precise measurement of heat flow out of a system.
- C) Energy stored in chemical bonds.
- D) Everything in the universe outside of the system.

28. An insulated device used to measure the absorption or release of heat in chemical or physical processes is known as a(n):

- A) Standard hydrogen electrode
- B) Calorimeter

- C) Barometer
- D) Cathode

29. The enthalpy change for a chemical equation exactly as it is written is referred to as the:

- A) Standard heat of formation
- B) Molar heat of vaporization

- C) Heat of reaction
- D) Heat capacity

30. Which term describes the amount of heat absorbed by one mole of a liquid as it vaporizes at a constant temperature?

- A) Molar heat of condensation (ΔH_{cond})
- B) Molar heat of fusion (ΔH_{fus})

- C) Molar heat of solution (ΔH_{soln})
- D) Molar heat of vaporization (ΔH_{vap})

31. The change in enthalpy that accompanies the formation of one mole of a compound from its elements with all substances in their standard states at 25°C is called the:

- A) Standard heat of formation (ΔH_f°)
- B) Heat of combustion

- C) Molar heat of solidification
- D) Thermochemical equation

32. Energy that transfers from one object to another because of a temperature difference between the objects is known as:

A) Heat (q)

B) Work

C) Specific heat

D) Chemical potential energy

33. What is the definition of an exothermic process?

A) A process that absorbs heat from the surroundings.

B) A process that releases heat to its surroundings.

C) A process where energy is neither created nor destroyed.

D) The precise measurement of heat flow out of a system.

34. The heat of reaction for the complete burning of one mole of a substance is the:

A) Molar heat of fusion

B) Heat of combustion

C) Molar heat of solution

D) Standard heat of formation

Match the definition in Column A with the correct term in Column B.

35. ____ Energy stored in chemical bonds.

36. ____ A process that absorbs heat from the surroundings.

37. ____ The amount of heat needed to increase the temperature of an object exactly 1°C.

38. ____ In any chemical or physical process, energy is neither created nor destroyed.

39. ____ Everything in the universe outside of the system.

40. ____ A part of the universe on which you focus your attention.

41. ____ The precise measurement of heat flow out of a system for chemical and physical processes.

42. ____ A chemical equation that includes the enthalpy change.

43. ____ The amount of heat released by one mole of a vapor as it condenses to a liquid at a constant temperature.

44. ____ The amount of heat absorbed by one mole of a solid substance as it melts to a liquid at a constant temperature.

45. ____ The amount of heat lost by one mole of a liquid as it solidifies at a constant temperature.

46. ____ The enthalpy change caused by the dissolution of one mole of a substance.

47. ____ If you add two or more thermochemical equations to give a final equation, then you also add the heats of reaction to give the final heat of reaction.

A. System

B. Surroundings

C. Chemical potential energy

D. Endothermic process

E. Heat capacity

F. Law of conservation of energy

G. Calorimetry

H. Thermochemical equation

I. Molar heat of condensation (ΔH_{cond})

J. Molar heat of fusion (ΔH_{fus})

K. Molar heat of solidification (ΔH_{solid})

L. Molar heat of solution (ΔH_{soln})

M. Hess's law of heat summation

Chapter 18

- 1. According to the collision theory, what condition is necessary for reactant particles to successfully form products?**
 - A. The particles must avoid physical contact to prevent degradation.
 - B. The particles must be heavily cooled before interaction.
 - C. The particles must be contained in a biological enzyme.
 - D. The particles must collide with a certain minimum amount of energy.
- 2. Why is activation energy often compared to a physical wall or barrier?**
 - e. Because it entirely prevents gases from mixing.
 - f. Because reactant particles must possess enough energy to "climb over" it before products can form.
 - g. Because it permanently traps the kinetic energy of the molecules.
 - h. Because it physically separates reactants from products in a container.
- 3. How does the presence of a catalyst affect the rate of a chemical reaction?**
 - E. It increases the rate by providing an alternative mechanism with a lower activation energy.
 - F. It decreases the rate by absorbing the kinetic energy of the reactants.
 - G. It increases the rate by raising the temperature of the entire system.
 - H. It has no effect on the rate but increases the product yield.
- 4. Which of the following statements regarding chemical reaction rates and biological systems is true?**
 - E. Increasing the temperature will consistently slow down a chemical reaction.
 - F. Once a reaction starts, particles no longer need to collide.
 - G. Enzymes function as biological catalysts.
 - H. Catalysts are always consumed during the first-order reaction step.
- 5. A gas stove requires a spark or flame to ignite but continues to burn spontaneously once lit. What does this indicate about the relationship between temperature and reaction rate?**
 - E. The flame is a catalyst that permanently alters the gas molecules.
 - F. Gas molecules are too cold to ever react without constant external heat.
 - G. The flame raises the temperature and energy of collisions to start the reaction, and the exothermic heat released maintains it.
 - H. Oxygen removes the activation energy barrier once the spark is introduced.
- 6. Which statement best defines a reversible reaction?**
 - E. A reaction where products completely disappear into a vacuum.
 - F. A reaction that must be restarted manually once it reaches equilibrium.
 - G. A reaction that only proceeds backwards after all reactants are consumed.
 - H. A chemical process where reactants are continually forming products, and products are continually forming reactants.
- 7. When a dynamic chemical equilibrium is established, what is the relationship between the rates of the forward and reverse reactions?**
 - E. The forward reaction is substantially faster.
 - F. The rates of the forward and reverse reactions are equal.
 - G. Both reactions completely halt.
 - H. The reverse reaction is substantially faster.
- 8. Using Le Chatelier's principle, what causes a carbonated drink to go flat when the container is left open?**
 - E. The system relieves the stress of the opened container by allowing the CO₂ gas above the liquid to escape, causing bubbles to leave the liquid to reestablish equilibrium.
 - F. The liquid expands, absorbing the dissolved gases into the water molecules.
 - G. The temperature of the room forces the equilibrium to favor solid precipitates.
 - H. Atmospheric pressure crushes the carbon dioxide molecules.
- 9. What is the correct expression for the equilibrium constant (K_{eq}) for the reaction: $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$?**

E. $K_{eq} = [PCl_5] / ([PCl_3] * [Cl_2])$

F. $K_{eq} = ([PCl_3] * [Cl_2]) / [PCl_5]$

G. $K_{eq} = [PCl_3] + [Cl_2] / [PCl_5]$

H. $K_{eq} = [PCl_5] * [PCl_3] * [Cl_2]$

10. If a reaction has an equilibrium constant (K_{eq}) of 1×10^2 , what does this suggest about the favorability of product formation?

E. Product formation is highly unfavorable.

F. The reaction is nonspontaneous.

G. The reaction will only produce reactants.

H. Product formation is favorable.

11. What is the correct solubility product expression for the salt Nickel(II) sulfide (NiS)?

E. $K_{sp} = [Ni^{2+}] / [S^{2-}]$

F. $K_{sp} = [Ni^{2+}] + [S^{2-}]$

G. $K_{sp} = [Ni^{2+}][S^{2-}]$

H. $K_{sp} = [NiS] / ([Ni^{2+}][S^{2-}])$

12. What does the solubility product constant (K_{sp}) mathematically represent?

E. The product of the ion concentrations raised to the power of their coefficients.

F. The total free energy released during dissolution.

G. The specific rate constant of the solid forming.

H. The exact mass of the solid dissolved in a supersaturated solution.

13. When using a solubility table, how do you identify the least soluble salt among a given list?

E. It will have the lowest K_{sp} value.

F. It will have the highest K_{sp} value.

G. It will have a K_{sp} value exactly equal to 1.

H. It will lack a common ion.

14. What effect does the addition of a common ion have on the solubility of another substance in the solution?

A. It heavily increases the solubility.

B. It has absolutely no effect.

C. It reverses the equilibrium completely to a gas phase.

D. It lowers the solubility.

15. How does the free energy of a reaction help determine its spontaneity?

E. A spontaneous reaction always has a positive free energy.

F. A spontaneous reaction always has a negative free energy.

G. A spontaneous reaction always has zero free energy.

H. Free energy cannot predict spontaneity; only enthalpy can.

16. What does the concept of entropy measure in a chemical system?

E. The total thermal energy of the system.

F. The degree of order or crystallization in a solid.

G. The measure of disorder in a system.

H. The kinetic energy of the specific rate constant.

17. If the products in a spontaneous process are more ordered than the reactants, how is this entropy change described?

E. It is considered a favorable entropy change.

F. It indicates the reaction must be zero-order.

G. It indicates the reaction is highly endothermic.

H. It is considered an unfavorable entropy change.

18. Which of the following systems possesses lower entropy?

E. A 50 mL sample of liquid water.

F. A 50 mL sample of solid ice.

- G. Both have identical entropy.
H. Entropy cannot be measured for water states.
19. **Predict the direction of the entropy change for the following synthesis reaction: $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$.**
E. Entropy decreases.
F. Entropy increases.
G. Entropy remains completely constant.
H. Entropy becomes unpredictable.
20. **Is it accurate to say that all spontaneous processes must be exothermic?**
e. Yes, because heat must always be released to do work.
f. Yes, endothermic processes can never happen on their own.
g. No, some endothermic processes are spontaneous due to a highly favorable increase in entropy.
h. No, because spontaneity is determined solely by the speed of the reaction.
21. **Steam condenses to liquid water at normal atmospheric pressure, which is a decrease in entropy. Why does this happen spontaneously?**
A. Condensation is a highly endothermic process.
B. The favorable exothermic change of the condensation process offsets the unfavorable entropy change.
C. The liquid water has higher activation energy.
D. Gases inherently prefer to be closely packed together regardless of temperature.
22. **What two thermodynamic factors work together to determine if a chemical reaction is spontaneous?**
A. Pressure change and volume change.
B. Activation energy and collision frequency.
C. Temperature and specific rate constant.
D. Enthalpy change and entropy change.
23. **What is the definition of a specific rate constant?**
A. A proportionality constant that relates the concentrations of reactants to the rate of reaction.
B. An expression relating the rate of a reaction to reactant concentrations.
C. A reaction where rate is proportional to only one reactant.
D. The constant that measures the disorder of a system.
24. **If half of a reactant in a first-order reaction disappears in 50 minutes, how much total time is required for the reaction to be 75% complete?**
A. 50 minutes
B. 75 minutes
C. 100 minutes
D. 150 minutes
25. **On a reaction progress curve showing the energy profile of a mechanism, what does the peak of the curve represent?**
A. The total energy of the products.
B. The baseline energy of the reactants.
C. The overall change in enthalpy.
D. The activation energy barrier.

Chapter 19

- According to the Arrhenius theory, how are acids defined?**
 - They are substances that accept a pair of electrons.
 - They are substances that ionize to give hydrogen ions in aqueous solution.
 - They are hydrogen-ion acceptors.
 - They are substances that donate a pair of electrons to form a covalent bond.
- What is the balanced equation for the dissociation of calcium hydroxide in water?**
 - $\text{Ca(OH)}_2 \rightarrow \text{Ca}^{2+} + 2\text{OH}^-$
 - $\text{Ca(OH)}_2 + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_3$
 - $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2$
 - $\text{Ca(OH)}_2 \rightarrow \text{Ca}^+ + \text{OH}^-$
- In the reaction $\text{HCl} + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{Cl}^-$, which reactant acts as the Brønsted-Lowry base (hydrogen-ion acceptor)?**
 - HCl
 - H_3O^+
 - Cl^-
 - H_2O
- Why is the Lewis theory considered more general than the Arrhenius and Brønsted-Lowry theories?**
 - It strictly applies only to neutral solutions.
 - It requires all acids to contain an oxygen atom.
 - It explains the behavior of compounds that act like bases without accepting hydrogen ions or act like acids without donating hydrogen ions.
 - It only applies to reactions involving a strong acid and a strong base.
- What are the concentrations of H^+ and OH^- in pure water at 25°C ?**
 - 1.0×10^{-14} M for both H^+ and OH^-
 - 7.0×10^{-7} M for both H^+ and OH^-
 - 1.0×10^{-7} M for both H^+ and OH^-
 - 1.0×10^{-1} M for both H^+ and OH^-
- If a solution has a hydrogen-ion concentration $[\text{H}^+]$ of 1.0×10^{-3} M, what is its pH and what type of solution is it?**
 - pH = 3.00; basic
 - pH = 11.00; acidic
 - pH = 3.00; acidic
 - pH = 11.00; basic
- What is the hydroxide-ion concentration $[\text{OH}^-]$ for a solution that has a pH of 9.00?**
 - 1.0×10^{-5} M
 - 1.0×10^{-9} M
 - 1.0×10^{-7} M
 - 1.0×10^{-14} M
- Which of the following best describes the acid dissociation constant (K_a) of a strong acid?**
 - A strong acid has a very small K_a because it is completely dissociated.
 - A strong acid has a very large K_a because it is completely dissociated.
 - K_a is exactly 1.0 for all strong acids.
 - Strong acids do not have a K_a because they do not dissolve in water.
- Assuming only one hydrogen is ionized, what is the correct equilibrium expression for the acid dissociation constant (K_a) of hydrocyanic acid (HCN)?**
 - $K_a = [\text{H}^+][\text{OH}^-]$
 - $K_a = [\text{HCN}] / ([\text{H}^+][\text{CN}^-])$
 - $K_a = [\text{H}^+][\text{CN}^-] / [\text{HCN}]$
 - $K_a = [\text{CN}^-] / [\text{H}^+]$

10. **What are the expected products of a general neutralization reaction?**
- A) A salt and water
 - B) A weak acid and a weak base
 - C) Hydrogen gas and a metallic oxide
 - D) A strong base and water
11. **During an acid-base titration, what signals that the end point has been reached?**
- A) A precipitate forms at the bottom of the flask.
 - B) The number of moles of hydrogen ions exactly equals the number of moles of hydroxide ions.
 - C) The indicator changes color.
 - D) The solution completely evaporates.
12. **What is the molarity of a sodium hydroxide (NaOH) solution if 40.0 mL of the solution is completely neutralized by 20.0 mL of a 0.50 M HCl solution?**
- A) 0.25 M
 - B) 0.50 M
 - C) 1.00 M
 - D) 0.15 M
13. **Why does an aqueous solution of sodium carbonate (Na_2CO_3) have a basic pH?**
- A) Weak-base cations donate protons to water, decreasing the pH.
 - B) Weak-acid anions accept protons from water, increasing the pH.
 - C) Strong-acid anions dissociate completely, lowering the pH.
 - D) The salt prevents the self-ionization of water entirely.
14. **Why can a buffered solution NOT absorb an unlimited amount of acid or base?**
- A) Buffers are only capable of neutralizing strong acids, not strong bases.
 - B) Eventually the buffer capacity is exceeded, causing the pH to change significantly.
 - C) The common ion effect causes the buffer to precipitate into a solid.
 - D) Buffers only function efficiently at a pH of exactly 7.00.
15. **What term is used for a substance that can act as both an acid and a base?**
- A) Diprotic
 - B) Neutral
 - C) Amphoteric
 - D) Conjugate
16. **Which term describes the negative ion formed when a water molecule loses a hydrogen ion?**
- A) Hydronium ion
 - B) Hydroxide ion
 - C) Conjugate acid
 - D) Lewis base
17. **What is an acid that contains three ionizable protons called?**
- A) Monoprotic acid
 - B) Diprotic acid
 - C) Triprotic acid
 - D) Strong acid
18. **Which term represents the product of the concentrations of hydrogen ions and hydroxide ions in water (1×10^{-14} at 25°C)?**
- A) Acid dissociation constant (K_a)
 - B) Base dissociation constant (K_b)
 - C) Ion-product constant for water (K_w)
 - D) Standard cell potential
19. **What do you call a measure of the amount of acid or base that may be added to a buffer solution before a significant change in pH occurs?**
- A) Buffer capacity

- B) Titration limit
- C) End point
- D) Equivalence point

Term

1. ____ Neutralization reaction
2. ____ Self-ionization
3. ____ Conjugate base
4. ____ Lewis acid
5. ____ Equivalence point
6. ____ Titration
7. ____ Strong base
8. ____ Salt hydrolysis
9. ____ Conjugate acid
10. ____ pH

Definitions:

- A.** A process in which the cations or anions of a dissociated salt accept hydrogen ions from water or donate hydrogen ions to water.
- B.** Process used to determine the concentration of a solution in which a solution of known concentration is added to a measured amount of unknown concentration until an indicator signals the end point.
- C.** Any substance that can accept a pair of electrons to form a covalent bond.
- D.** The point in a titration where the number of moles of hydrogen ions equals the number of moles of hydroxide ions.
- E.** A number used to denote the hydrogen-ion concentration, or acidity, of a solution; the negative logarithm of the hydrogen-ion concentration.
- F.** The particle formed when a base gains a hydrogen ion.
- G.** A base that completely dissociates into metal ions and hydroxide ions in aqueous solution.
- H.** A reaction in which an acid and a base react in an aqueous solution to produce a salt and water.
- I.** A term describing the reaction in which two water molecules react to produce ions.
- J.** The particle that remains when an acid has donated a hydrogen ion.

Chapter 20

1. What chemical process must always accompany an oxidation process?

- A. Combustion
- B. Neutralization
- C. Reduction
- D. Sublimation

2. What happens to a reducing agent during a redox reaction?

- A. The reducing agent is oxidized.
- B. The reducing agent is reduced.
- C. It gains electrons.
- D. Its oxidation number decreases.

3. Balance the redox equation and identify whether the first substance was oxidized or reduced: $\text{Mg(s)} + \text{O}_2\text{(g)} \rightarrow \text{MgO(s)}$

- A. $\text{Mg(s)} + \text{O}_2\text{(g)} \rightarrow \text{MgO}_2\text{(s)}$; magnesium is reduced
- B. $2\text{Mg(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{MgO(s)}$; magnesium is oxidized
- C. $2\text{Mg(s)} + 2\text{O}_2\text{(g)} \rightarrow 2\text{MgO(s)}$; oxygen is oxidized
- D. $\text{Mg(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{MgO(s)}$; magnesium is oxidized

4. Identify the following process as either oxidation or reduction: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$

- A. Oxidation
- B. Reduction
- C. Both oxidation and reduction
- D. Neither oxidation nor reduction

5. Which of the following would most likely be a reducing agent? (Hint: Think in terms of tendencies to lose or gain electrons based on electronegativity.)

- A. F_2
- B. O_2
- C. Na
- D. Cl_2

6. Determine which reactant is oxidized, which is reduced, and identify the agents for the following reaction: $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$

- A. Zn is oxidized (reducing agent); H^+ is reduced (oxidizing agent).
- B. Zn is reduced (oxidizing agent); H^+ is oxidized (reducing agent).
- C. Cl^- is oxidized (reducing agent); Zn is reduced (oxidizing agent).
- D. H^+ is oxidized (oxidizing agent); Zn is reduced (reducing agent).

7. Which of the following best defines an oxidation number?

- A. The exact number of electrons an atom possesses in a molecule.
- B. The total number of bonds an atom makes in a compound.
- C. The charge an atom would have if the electrons in each bond were assigned to the atoms of the more electronegative element.
- D. The number of protons minus the number of neutrons.

8. Which of these statements is false?

- A. The sum of the oxidation numbers of the atoms in a polyatomic ion must equal the charge of the ion.
- B. The oxidation number of an uncombined element is zero.
- C. The oxidation number of oxygen in a compound or polyatomic ion is almost always +2.
- D. Fluorine always has an oxidation number of -1 in its compounds.

9. Determine the oxidation number of the transition metal atom in KMnO_4 .

- A. +1
- B. +2
- C. +5
- D. +7

10. Assign oxidation numbers to the atoms in the sulfate ion (SO_4^{2-}).

- A. S = +6; O = -2
- B. S = +4; O = -2
- C. S = +8; O = -2
- D. S = -2; O = +1

11. Use the changes in oxidation numbers to identify which atoms are oxidized and which are reduced in the reaction: $2\text{Fe}^{3+}(\text{aq}) + \text{Sn}^{2+}(\text{aq}) \rightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{Sn}^{4+}(\text{aq})$

- A. Fe^{3+} is oxidized; Sn^{2+} is reduced.
- B. Sn^{2+} is oxidized; Fe^{3+} is reduced.
- C. Both Fe^{3+} and Sn^{2+} are oxidized.
- D. Both Fe^{3+} and Sn^{2+} are reduced.

12. Balance the following redox equation: $\text{Na}(\text{s}) + \text{Cl}_2(\text{g}) \rightarrow \text{NaCl}(\text{s})$

- A. $\text{Na}(\text{s}) + \text{Cl}_2(\text{g}) \rightarrow \text{NaCl}_2(\text{s})$
- B. $2\text{Na}(\text{s}) + 2\text{Cl}_2(\text{g}) \rightarrow 2\text{NaCl}(\text{s})$
- C. $2\text{Na}(\text{s}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{NaCl}(\text{s})$
- D. $\text{Na}(\text{s}) + 2\text{Cl}_2(\text{g}) \rightarrow \text{NaCl}_4(\text{s})$

13. Which of the following unbalanced equations represents a redox reaction?

- A. $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- B. $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$
- C. $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
- D. $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{NaNO}_3(\text{aq})$

14. Use the half-reaction method to identify the balanced ionic equation for the following reaction in a basic solution: $\text{CN}^-(\text{aq}) + \text{MnO}_4^-(\text{aq}) \rightarrow \text{CNO}^-(\text{aq}) + \text{MnO}_2(\text{s})$

- A. $3\text{CN}^-(\text{aq}) + 2\text{MnO}_4^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow 3\text{CNO}^-(\text{aq}) + 2\text{MnO}_2(\text{s}) + 2\text{OH}^-(\text{aq})$
- B. $\text{CN}^-(\text{aq}) + \text{MnO}_4^-(\text{aq}) \rightarrow \text{CNO}^-(\text{aq}) + \text{MnO}_2(\text{s})$
- C. $3\text{CN}^-(\text{aq}) + 2\text{MnO}_4^-(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow 3\text{CNO}^-(\text{aq}) + 2\text{MnO}_2(\text{s}) + \text{H}_2\text{O}(\text{l})$
- D. $2\text{CN}^-(\text{aq}) + 3\text{MnO}_4^-(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{CNO}^-(\text{aq}) + 3\text{MnO}_2(\text{s}) + 6\text{OH}^-(\text{aq})$

15. Which term describes a process that involves a complete or partial gain of electrons or the loss of oxygen?

- A. Oxidation
- B. Reduction
- C. Electrolysis
- D. Sublimation

16. What do we call the substance in a redox reaction that accepts electrons?

- A. Reducing agent
- B. Catalyst
- C. Oxidizing agent
- D. Common ion

17. What is an equation showing either the oxidation or the reduction that takes place in a redox reaction?

- A. Reaction mechanism
- B. Net ionic equation
- C. Thermochemical equation
- D. Half-reaction

18. Which method involves balancing a redox equation by comparing the increases and decreases in oxidation numbers?

- A. Half-reaction method
- B. Oxidation-number-change method
- C. Specific rate constant method
- D. Common ion effect

Instructions: Match the Chapter 20 term on the left with its exact definition from the glossary on the right.

Terms:

- 50. ____ Oxidation
- 51. ____ Oxidation-reduction reaction
- 52. ____ Oxidizing agent
- 53. ____ Redox reaction
- 54. ____ Reducing agent
- 55. ____ Reduction
- 56. ____ Oxidation number
- 57. ____ Half-reaction
- 58. ____ Half-reaction method
- 59. ____ Oxidation-number-change method

Definitions:

- A.** Another name for an oxidation-reduction reaction.
- B.** An equation showing either the oxidation or the reduction that takes place in a redox reaction.
- C.** A process that involves complete or partial loss of electrons or a gain of oxygen; it results in an increase in the oxidation number of an atom.
- D.** A method of balancing a redox equation by comparing the increases and decreases in oxidation numbers.
- E.** The substance in a redox reaction that donates electrons; in the reaction, this substance is oxidized.
- F.** A reaction that involves the transfer of electrons between reactants.
- G.** A positive or negative number assigned to an atom to indicate its degree of oxidation or reduction; the number for an uncombined element is zero.
- H.** A process that involves a complete or partial gain of electrons or the loss of oxygen; it results in a decrease in the oxidation number of an atom.
- I.** The substance in a redox reaction that accepts electrons; in the reaction, this substance is reduced.
- J.** A method of balancing a redox equation by balancing the oxidation and reduction components separately before combining them into a balanced redox equation.

Chapter 21

- If a piece of solid silver is submerged in a solution of zinc nitrate, what is the expected outcome based on their relative reactivities?**
 - Silver will be oxidized.
 - No reaction will occur.
 - Zinc metal will precipitate.
 - Hydrogen gas will be released.
- Which of the following metals has the greatest tendency to undergo oxidation compared to the others?**
 - Copper (Cu)
 - Magnesium (Mg)
 - Silver (Ag)
 - Mercury (Hg)
- In any functioning electrochemical cell, the process of gaining electrons strictly takes place at the:** A. Cathode B. Anode C. Salt bridge D. Voltmeter
- What is the primary purpose of a salt bridge in a standard voltaic cell?**
 - To supply the activation energy required to start the redox reaction.
 - To provide a pathway for electrons to flow between the two electrodes.
 - To maintain electrical neutrality by allowing ions to flow without mixing the solutions.
 - To prevent the migration of ions between the half-cells completely.
- In a standard dry cell battery commonly used in household items, which materials typically serve as the anode and cathode, respectively?**
 - Carbon/graphite (anode) and manganese dioxide (cathode)
 - Copper (anode) and zinc (cathode)
 - Lead (anode) and lead(IV) oxide (cathode)
 - Zinc (anode) and carbon/graphite (cathode)
- As a traditional lead storage car battery discharges, why does the specific gravity (density) of its electrolyte solution drop?**
 - Sulfuric acid is consumed and water is produced, leaving a liquid with a lower density.
 - Solid lead dissolves into the solution, decreasing its mass.
 - The lead sulfate produced is highly soluble and lighter than sulfuric acid.
 - Hydrogen and oxygen gases bubble out, removing mass from the liquid.
- Which shorthand notation correctly outlines the active solid components of a discharging lead-acid storage battery?**
 - $\text{Zn(s)} \mid \text{Zn}^{2+}(\text{aq}) \parallel \text{Cu}^{2+}(\text{aq}) \mid \text{Cu(s)}$
 - $\text{Pb(s)} \mid \text{PbSO}_4(\text{s}) \parallel \text{PbO}_2(\text{s}) \mid \text{PbSO}_4(\text{s})$
 - $\text{PbO}_2(\text{s}) \mid \text{Pb(s)} \parallel \text{PbSO}_4(\text{s}) \mid \text{H}_2\text{O(l)}$
 - $\text{Pt(s)} \mid \text{H}_2(\text{g}) \parallel \text{O}_2(\text{g}) \mid \text{H}_2\text{O(l)}$
- Despite producing nearly zero emissions, hydrogen fuel cells are not yet the dominant power source for commercial electricity. What is the primary reason for this?**
 - They are currently less economical to operate than conventional electrical generation methods.
 - They produce highly toxic solid waste that is hard to dispose of.
 - The chemical reactions involved violate the laws of thermodynamics.
 - They must be continuously recharged from a wall outlet to function.
- Which of the following represents a major advantage of a hydrogen fuel cell compared to a traditional lead storage battery?**
 - It requires frequent recharging by reversing the electrical current.
 - It is much heavier, providing better stability for vehicles.
 - It relies on a concentrated sulfuric acid electrolyte, which is cheap.
 - It does not need recharging as long as fuel is supplied, and its primary byproduct is water.

10. **How do chemists know that the standard reduction potential of the standard hydrogen electrode is exactly 0.00 V?**
- A. It was measured using a perfectly frictionless voltmeter.
 - B. It was arbitrarily assigned this value to serve as a universal reference point.
 - C. It was calculated using the ideal gas law at 298 K.
 - D. It is the exact amount of energy released when a single hydrogen bond breaks.
11. **How is the electric potential of a cell best described?**
- A. The total number of electrons flowing through the wire per second.
 - B. The physical distance between the anode and cathode.
 - C. A measure of the cell's ability to produce an electric current.
 - D. The sum of the masses of the two electrodes.
12. **When comparing an activity series table to a table of standard reduction potentials, what relationship exists regarding the order of elements?**
- A. They follow the same relative sequence because both rank elements by their tendency to undergo redox processes.
 - B. There is no correlation between an element's activity and its standard potential.
 - C. They are exact opposites because one measures oxidation and the other measures reduction.
 - D. The tables only match for nonmetals, while metals are arranged randomly.
13. **If you needed to experimentally determine the standard reduction potential of the aluminum half-cell, what procedure would you use?**
- A. Weigh the metal before and after it reacts with an acid.
 - B. Dissolve the metal in distilled water and measure the change in pH.
 - C. Connect the aluminum half-cell to a standard hydrogen electrode and read the voltage on a voltmeter.
 - D. Heat the metal until it melts and record the temperature at standard pressure.
14. **A proposed electrochemical reaction has a calculated standard cell potential of -1.24 V. What does this value indicate about the reaction?**
- A. It is spontaneous and will proceed without an external power source.
 - B. It is nonspontaneous and requires an input of electrical energy to occur.
 - C. It is at perfect equilibrium and will produce an alternating current.
 - D. It is spontaneous only at temperatures above absolute zero.
15. **If a voltaic cell consists of a tin half-cell (reduction potential = -0.14 V) and a silver half-cell (reduction potential = +0.80 V), what is the overall standard cell potential?**
- A. -0.94 V B. -0.66 V C. +0.94 V D. +0.66 V
16. **Why is alternating current (AC) unsuitable for electroplating a metal object?**
- A. AC causes the electrolyte solution to boil instantly.
 - B. AC changes direction constantly, meaning the current would deposit metal and then immediately remove it.
 - C. AC flows in one direction only, which prevents the metal from depositing.
 - D. AC does not carry enough voltage to overcome the activation energy.
17. **To properly electroplate a piece of jewelry with pure silver, how should the electrolytic cell be set up?**
- A. The jewelry should act as the cathode, submerged in an electrolyte containing silver cyanide.
 - B. The jewelry should be the anode, and silver ions should be in the solution.
 - C. The jewelry should be attached to the salt bridge to attract silver atoms.
 - D. The jewelry and a piece of solid silver should be rubbed together while applying a current.
18. **Which of the following correctly represents the overall reaction for the electrolytic decomposition of liquid water?**
- A. $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$
 - B. $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}^+(\text{aq}) + \text{OH}^-(\text{aq})$
 - C. $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$
 - D. $4\text{H}_2\text{O}(\text{l}) \rightarrow 4\text{H}_2(\text{g}) + 2\text{O}_2(\text{g}) + 4\text{e}^-$

19. **During the electrolysis of water in a dual-tube apparatus, how can you visually identify which tube contains the hydrogen gas?**
- A. It produces gas at half the volume of the oxygen gas.
 - B. It forms at the positive anode.
 - C. It produces gas at twice the volume of the oxygen gas.
 - D. The hydrogen gas is bright yellow, making it visually obvious.
20. **When performing the electrolysis of an aqueous sodium chloride solution (brine), hydrogen and chlorine gases are evolved. In what molecular ratio should these two gases form according to the balanced chemical equation?**
- A. 3 molecules of hydrogen for every 1 of chlorine (3:1)
 - B. 1 molecule of hydrogen for every 1 of chlorine (1:1)
 - C. 1 molecule of hydrogen for every 2 of chlorine (1:2)
 - D. 2 molecules of hydrogen for every 1 of chlorine (2:1)
21. **What is the fundamental difference in energy conversion between a voltaic cell and an electrolytic cell?**
- A. Voltaic cells convert chemical energy into electrical energy; electrolytic cells use electrical energy to cause a nonspontaneous chemical reaction.
 - B. Voltaic cells convert electrical energy into chemical energy; electrolytic cells convert chemical into electrical.
 - C. Voltaic cells convert heat into chemical energy; electrolytic cells do the reverse.
 - D. Both cells perform the exact same function, but electrolytic cells use a salt bridge.
22. **Why must electrochemical potentials always be measured using a pair of connected half-cells rather than a single, isolated half-cell?**
- A. Oxidation or reduction cannot occur in isolation; electrons must have both a source and a destination to create a measurable flow.
 - B. A single half-cell does not contain enough liquid to register on a voltmeter.
 - C. Voltmeters are designed with two wires that physically cannot touch the same solution.
 - D. An isolated half-cell produces too much voltage and would break standard measuring equipment.

Chapter 25

1. Which of the following describes the change in an atom's nucleus when it loses an alpha particle?

- A. The mass number is unchanged, and the atomic number increases by 1.
- B. The mass number decreases by 4, and the atomic number decreases by 2.
- C. The mass number and atomic number are both unchanged.
- D. The mass number decreases by 2, and the atomic number decreases by 4.

2. What is the definition of a radioisotope?

- A. An isotope that is radioactive.
- B. An isotope that occurs only in nature.
- C. An isotope that has the same atomic mass as other isotopes.
- D. An isotope that does not emit radiation.

3. Why do nuclear power plants remove spent fuel rods from a reactor core?

- A. Because they contain only non-fissionable isotopes.
- B. Because the reactor cannot maintain a constant electrical output with them.
- C. Because they become too cold to generate electricity.
- D. Because they are no longer radioactive.

4. What is the primary reason nuclear fusion is not currently used to generate electrical power?

- A. It produces too much radioactive waste compared to fission.
- B. It requires extremely high temperatures, making it difficult to start or contain the reaction.
- C. It does not release enough energy to be useful.
- D. It is only possible with elements that are too rare to find.

5. How is the process of nuclear fission described?

- A. The combining of small nuclei to release energy.
- B. The bombardment of nuclei with neutrons, causing them to split into two fragments and release energy.
- C. The process of removing electrons from atoms to create ions.
- D. The natural decay of an element into a different element without external bombardment.

6. What is the difference between natural and artificial radioactivity?

- A. Natural radioactivity involves fission, while artificial involves fusion.
- B. Natural radioactivity comes from elements in nature, while artificial radioactivity comes from elements created in nuclear reactors and accelerators.
- C. Artificial radioactivity is not actually radioactive.
- D. Natural radioactivity only occurs in transuranium elements.

7. Why is x-ray and radioisotope radiation referred to as "ionizing radiation"?

- A. Because it has enough energy to remove electrons from the atoms it hits.
- B. Because it creates new elements through transmutation.
- C. Because it is only emitted during nuclear fission.
- D. Because it is visible to the human eye.

8. Alpha Decay of Radium-226 The disintegration of the radioisotope radium-226 produces an isotope of radon and an alpha particle. Which of the following is the balanced nuclear equation?

- A. ${}^{226}_{88}\text{Ra} \rightarrow {}^{226}_{86}\text{Rn} + {}^4_2\text{He}$
- B. ${}^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + {}^4_2\text{He}$
- C. ${}^{226}_{88}\text{Ra} \rightarrow {}^{222}_{88}\text{Rn} + {}^4_2\text{He}$
- D. ${}^{226}_{88}\text{Ra} \rightarrow {}^{224}_{86}\text{Rn} + {}^2_2\text{He}$

9. Beta Decay of Carbon-14 Carbon-14 is a beta emitter. When it undergoes beta decay, what is the correct balanced equation for the production of nitrogen?

- A. ${}^{14}_6\text{C} \rightarrow {}^{14}_7\text{N} + {}^0_{-1}\text{e}$
- B. ${}^{14}_6\text{C} \rightarrow {}^{14}_5\text{N} + {}^0_{-1}\text{e}$
- C. ${}^{14}_6\text{C} \rightarrow {}^{13}_7\text{N} + {}^1_0\text{n}$
- D. ${}^{14}_6\text{C} \rightarrow {}^{14}_6\text{N} + {}^0_0\gamma$

10. Effects of Gamma Radiation When a nucleus emits only a gamma ray, how are the atomic number and mass number of the nucleus affected?

- A. The mass number decreases by 4, and the atomic number decreases by 2
- B. The mass number increases by 1, and the atomic number remains unchanged
- C. Both the mass number and the atomic number are unchanged
- D. The atomic number decreases by 1, and the mass number remains unchanged.

Term

- 11. _____ Fission
- 12. _____ Fusion
- 13. _____ Transmutation
- 14. _____ Beta particle
- 15. _____ Gamma ray

Defintiion

- A. The conversion of an atom of one element to an atom of another.
- B. The splitting of a nucleus into smaller fragments, releasing energy.
- C. The process of combining nuclei to produce a nucleus of greater mass.
- D. An electron resulting from the breaking apart of neutrons in an atom.
- E. A high-energy photon emitted by a radioisotope.